Towards Assumption-Based Argumentation Mining in Hotel Reviews

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Abstract. Argumentation mining (or argument mining) is the research area aiming at extracting human arguments and their relations (primarily) from text, with the ultimate goal to understand humans' arguments in argumentative texts. Nonetheless, the progresses thus far often ignore the computational models of structured argumentation formalisms which are well studied in the knowledge representation and reasoning (KRR) community. This means that there is a still big research gap if ones would like to automatically construct argumentation-based knowledgebase systems from text. In this paper, we investigate this problem and formally define important tasks for *argument mining* from the views of machine learning and KRR. Due to current progresses of KRR, there could be a plethora of tasks depending on targeted structured argumentation frameworks. This paper focuses on a well-known *assumption-based argumentation* (ABA) by defining relevant tasks for mining ABA from text and then accordingly build an annotated corpus in hotel reviews from Booking.com. Finally, we evaluate the in-context learning capabilities of large language models (i.e. GPT-4o) for an ABA's construction based on our annotated hotel reviews' corpus, demonstrating the proposed framework is promising in practice.

Keywords: Argumentation System · Automatic Knowledge Base Construction · Text Corpus · Machine Learning Pipeline

1 Introduction

We have an academic dream that one day machine learning (ML) researchers and knowledge representation and reasoning (KRR) researchers can sit down together at the table of a joint panel and discuss on how to progress and realize

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'automated knowledge representation' systems from unstructured data such as natural language text. This similar dream was once stated in the work of [32, p. 5427] in the context of natural language processing (NLP) for ML and argumentation for KRR, which gives a nicely comprehensive survey of *argumentation mining* (or, shortly, argument mining) works.

Note that argument mining [18,32] is the research area aiming at automatically identifying and extracting the structure of inference and reasoning expressed as arguments presented in natural language. Grasping arguments' structure helps identify not only the stances that people take but also the reasons behind their opinions. This understanding offers valuable insights across various fields (see [32] for a review). Approaches towards this aim of argument mining was firstly introduced around 2010, showing techniques of mining (different connotations of) arguments from natural language. For example, [27] introduced a notion called *argumentative zoning* in scientific articles. [21] proposed an approach to detect legal arguments. Since these pioneering methods, the necessity for automating the extraction of arguments and their interconnections from natural language text was highlighted, though it received only limited attention. Currently, argument mining is well recognized as a sub-area of NLP that exploits philosophical/formal argumentation to deal with the automated representation, evaluation, and generation of arguments. For instance, [20] considered complex rhetorical moves in attacks or the presuppositions or value judgments in argumentation. Another example, [15] paid an attention to argumentative feedback systems in order to assist learners in improving the quality of their arguments.

Despite big progresses in argument mining, it is still relatively under-explored on how to deal with the automated representation, evaluation, and generation of arguments regarding the KRR's principles. Indeed, the NLP techniques disclosed in argumentation mining are not sufficient to mine arguments in computational forms. This problem could hinder diverse applications of formal argumentation, such as, Dung's argumentation [11], bipolar argumentation [4], and assumption-based argumentation (ABA) [2], in text documents. Note that there exist several studies aimed to fill this gap; however, formal argumentation formalisms are not still comprehensively explored. Exceptional works include [19] who developed an annotated text corpus for mining the bipolar argumentation, and [7] who studied NLP models to mine bipolar arguments.

To bridge the gap between computational models in KRR and ML, this paper specifically coins out a novel sub-field called *assumption-based argumentation mining* (ABA mining) in order to raise awareness on the lack of investigating NLP techniques for this purpose. Our main contributions are twofold as follows: **(1)** We introduce a novel annotation framework for mining ABA representation from text (cf. Section 3). **(2)** Accordingly, we develop a novel annotated corpus based on our framework in a context of Hotel Reviews [17] (cf. Section 4).

The remaining sections are structured as follows. Section 2 presents the basics of ABA. Section 5 shows our findings towards our ABA mining tasks by leveraging the in-context learning capabilities of large language models (i.e. GPT-4o). We relate the work in Sect. 7 and conclude in Sect. 8.

2 Preliminary: Assumption-Based Argumentation

Assumption-based argumentation (ABA) frameworks are 4-tuples $(\mathcal{L}, \mathcal{R}, \mathcal{A}, \overline{})$, where

- $(\mathcal{L}, \mathcal{R})$ is any deductive system with a language $\mathcal{L}$ and a set of inference rules of the form $l_0 \leftarrow l_1, \ldots l_n$ ($n \geq 0$ and $l_i \in \mathcal{L}$),
- $\mathcal{A} \subseteq \mathcal{L}$ represents a non-empty set of assumptions, and
- $\overline{} : \mathcal{A} \rightarrow 2^{\mathcal{L}} \setminus \emptyset$ is a total mapping such that $\overline{\alpha}$ represents a non-empty set of *contraries* of α.

Note that, in the original ABA framework [14], $\overline{}$ is a total mapping from $\mathcal{A}$ into $\mathcal{L}$; each assumption has a single contrary. This paper adopts an extension proposed in [14] as it better reflects the characteristics of raw data in our application domain, i.e., Hotel Reviews Booking.com (cf. Section 3).

For each rule $r \in \mathcal{R}$ of the form $l_0 \leftarrow l_1, \ldots l_n$, l_0 and the set $\{l_1, \ldots, l_n\}$ are referred respectively as the head and the body of r and denoted by $hd(r)$, $bd(r)$. We restrict ourselves to the flat ABA framework [2,14] such that the head of any inference rule cannot be an assumption, i.e., for any rule $r \in \mathcal{R}$, $hd(r) \notin \mathcal{A}$.

Any argument in ABA is in a tree structures represented by a deduction of a conclusion $c \in \mathcal{L}$ derived from a set of assumptions S with c at the root and S at the leaves. ABA is an instance of Dung's abstract argumentation (AA), and vice versa [9,10,28]. Hence, all semantic notions for determining the acceptability of arguments in AA are applicable to arguments in ABA, offering implementation methodologies for various application domains such as legal reasoning [12], medical reasoning [8], and decision theory [1]. Hence, it would be more worthwhile to have an automated labelling system to transform textual description of any application domain into such an ABA framework.

3 An Annotation Framework for ABA Mining in Text

To promote industrial impacts of ABA, we introduce a novel 'annotation framework' for ABA mining from text. In our framework, literals in $\mathcal{L}$ are represented by propositions due to its simplicity of developing machine learning-based mining platforms. For example, 'very helpful staff' in the raw data is converted into proposition 'helpful_staff' and 'option to have a good breakfast within the hotel save your time' is to proposition 'good_breakfast' (cf. Figure 1).

Since any ABA framework is the quadruple, we propose that any ABA mining platform must handle the following two problems which constitutes a sequential workflow of any ABA mining system.

1. *Inference rule mining*: requires obtaining a set of (inference) rules given that a document. Clearly, obtaining a rules' set allows to realize the components $(\mathcal{L}, \mathcal{R})$ and component $\mathcal{A}$ of the corresponding ABA framework.
2. *Contrary mining*: requires obtaining a contraries' set of any assumption a in $\mathcal{A}$ given that a document and the corresponding assumption set $\mathcal{A}$.

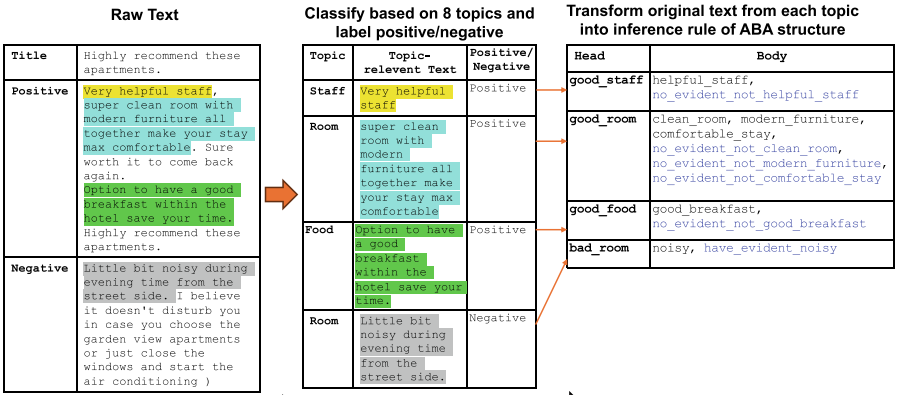

Fig. 1. Transforming raw hotel reviews in Booking.com into an ABA framework.

There could be various possible ways to address the above tasks, depending on the characteristics of raw data. In this paper, we aim at providing concrete methodologies including the annotated dataset and benchmark results using the state-of-the-art generative model (i.e. GPT-4o) for automatically constructing an ABA knowledge base system.

3.1 Raw Data: Hotel Reviews Booking.com

Our analysis is based on Hotel Reviews Booking.com in kaggle [17]. The dataset was scraped from La Veranda Hotel on Booking.com to contain 1,500 customer reviews for a hotel in Larnaca-Cyprus. These reviews provide the experiences of guests who stayed at the property during its first year of operation. Figure 1 (Raw Text) depicts example snapshots of the raw dataset. It consists of three parts: a title, a positive comment (labelled by 'positive'), and a negative comment (labelled by 'negative'); all parts were provided by the hotel's guests. Based on our analysis, the titles do not give valuable insights, compared to the others. Hence, we propose to mine the language and inference rules $(\mathcal{L}, \mathcal{R})$ from the positive and negative comments only.

The ABA-based corpus annotating framework proposed in this work has three annotating steps corresponding to each defined task (cf. Figure 2). Given any sentence in the raw data, the first step (corresponding to Task 1) is to annotate the *zoning* of each relevant text in the original sentence and label its corresponding *topic* and its *sentiment* (either 'positive' or 'negative'). Next, corresponding to Task 2, given a set of triples of the form (*topic*, *topic-relevant text*, *sentiment*), the second step is to annotate the *head* and the *body* of each corresponding inference rule. Clearly, the results of this annotating step yield $(\mathcal{L}, \mathcal{R})$ of ABA. The last step, which is corresponding to Task 3, is to label the *contraries* for each assumption in $\mathcal{A}$. This framework is analyzed based on the hotel review dataset mentioned in the above. Nevertheless, this framework is general

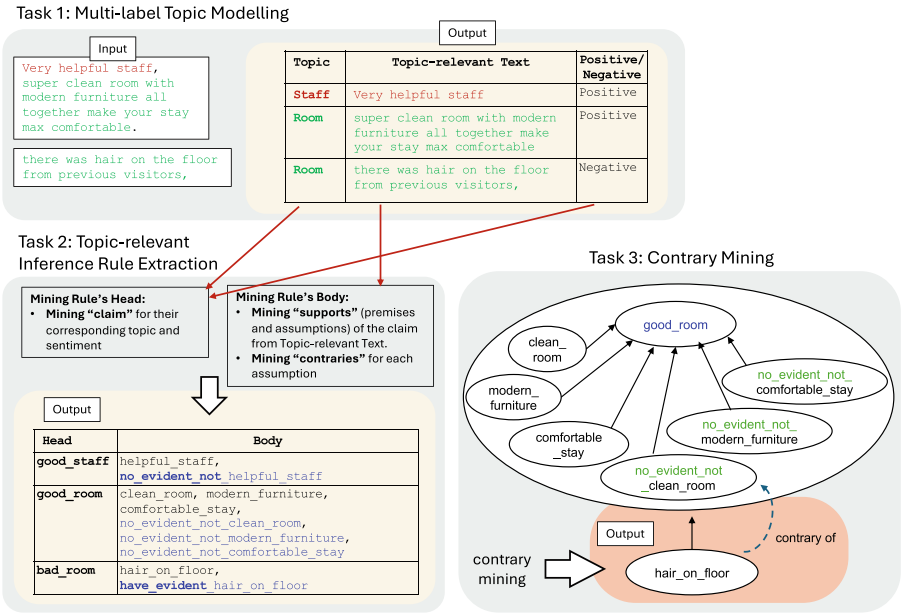

Fig. 2. Our ABA-based corpus annotating framework.

in a sense that it can be applied to other domains if the datasets include the topics, the positive comments, and the negative comments.

Figure 2 also shows the three corresponding tasks involved in our annotating framework: **(Task 1)** multi-label topic modelling, **(Task 2)** topic-relevant inference rule extraction, **(Task 3)** contrary mining. Indeed, Task 1 extracts (multiple) topics from an input document. Task 2 seeks for inference rules for all topics in the document. Task 3 extracts the contraries' set from the document. This design is agreed by two argumentation and one NLP researchers to ensure the applicability of any ABA reasoning engines (e.g. [14]) on the extracted ABA framework using the corpus. We detail all the tasks in the following subsections.

3.2 Task 1: Multi-label Document Typesetting

Definition 1. *Let T be a set of topics, D be a set of documents, Txt be a set of texts in D, and $S := \{0, 1\}$ be a set of boolean sentiments. We say that any* multi-label topic modelling *m is a function that maps from D into $D^{\downarrow}$, where $D^{\downarrow} \subseteq (T \times Txt \times S) \setminus \emptyset$ denotes a multi-label typeset document.*

Definition 1 outlines input-output representations for learning to label (multiple) topics in any input text. Learning such a multi-label topic modelling can be handled by various techniques, such as, unsupervised topic models and multi-label classification [3]. In this regard, our formal definition is general in a

sense that it does not enforce any implementation techniques, but rather outlines how an annotated corpus should be created in order to enable the development of efficient models for mining ABA structure. Figure 2 (cf. Task 1) shows examples of the annotations, which can be seen as, for any document in D, there is a mapped triple consisting of a topic, its topic-relevant text, and its sentiment.

3.3 Task 2: Topic-Relevant Inference Rule Extraction

Given a set of topics, their corresponding relevant text, and the sentiments, we further extract inference rules and assumptions from the original document.

Definition 2. *Let $D^{\downarrow}$ be a multi-label typeset document, $\mathcal{R}$ be a set of inference rules, and $\mathcal{A}$ be a set of assumptions. We say that any* topic-relevant inference rule extraction *m is a function that maps from $D^{\downarrow}$ into $\mathcal{R} \times A$.*

Like Definition 1, Definition 2 also outlines a general annotation framework for learning and evaluating inference rule extraction models. This work proposes to deal with this task by addressing two subtasks: **(1)** mining the rule's head and **(2)** mining the rule's body. Regarding the first subtask, we exploit sentiments of topics provided in $D^{\downarrow}$. If a sentiment is positive, the head is set to the topic's name prefixed by 'good_'. Otherwise, the topic's name is prefixed by 'bad_'. Furthermore, mining the rule's body can be viewed as a rewriting problem where a natural language text is rewritten as a sequence of semantic-identical premises with corresponding assumptions. These assumptions enables any deduction for claims to be 'undercut' as in the sense of ABA.

To ease the annotation, we use a rule-based procedure for generating assumptions. Indeed, if the sentiment is positive, each premise of the rule is a newly fresh proposition prefixed with 'no_evident_not_'. Otherwise, each premise of the rule is a newly fresh literal that prefixes its name with 'have_evident_'. Obviously, solving these two subtasks yield the output's representation as defined.

4 Corpus Creation for Hotel Reviews Booking.com

This section describes our data creation method of our annotated corpus for learning to mine an ABA framework from text, called *ABA-HotelReviews*, following the annotation framework outlined in Sect. 3. By following the proposed annotating framework, ABA-HotelReviews enables to learn machine learning models for extracting an ABA framework from the hotel reviews dataset.

Our main goal is to create an annotated corpus according to the theoretical framework of assumption-based argumentation. Thus, we start our process from investigating existing crawled datasets that requires least pre-processing for our annotation steps (cf. Section 3) based on the ABA's formalism. First, we select Hotel Reviews Booking.com in kaggle [17] (cf. Subsection 3.1) because the reviews contain arguments from customers by the domain's nature, and then proceed to creating different corpora for all the tasks proposed in Sect. 3. We use three domain experts to ensure the correctness of the annotated data for

each task. All of them are familiar with knowledge representation and argumentation. We detail how the corpora for Task 1 and Task 2 are created below. We leave the creation for contrary mining as our future task. Figure 7 illustrates a visualization of the extracted ABA from the target dataset.

4.1 Creation of Multi-label Topic in Hotel Reviews

The scraped dataset from the La Veranda hotel in Booking.com contains 11 columns: Title, Positive Review, Negative Review, Score, Guest Name, Guest Country, Room Type, Number of Nights, Visit Date, Group Type, and Property Response. Our primary objective is to understand the arguments within this dataset. Therefore, we focus solely on the columns Positive Review and Negative Review, which detail what guests liked and disliked about the hotel.

Below is the pre-processing steps for preparing a multi-label topic dataset of La Veranda Hotel reviews:

1. **Remove unused data:** Since not all guests provided comments, some only rated a score (a satisfaction rating out of 10), then we removed rows without either a positive review or a negative review, resulting in 785 usable reviews out of 1500 total customer reviews (see Table 3).
2. **Determine topics:** We examined the dataset and agreed on classifying that each argumentative text is classified into eight potential topics: (1) room, (2) staff, (3) location, (4) food, (5) price, (6) facility, (7) check-in, (8) check-out. If the texts are outside these topics, they are classified as noises.
3. **Labeling rules:** For each review, we assigned a relevant topic to each phrase or sentence. Then, we corrected obvious typos appeared in the phrases and sentences, such as '5 lins away from the airport' to '5 mins away from the airport'. Sometimes, guests did not strictly write all positive experiences in the positive review and, similarly, all negative experiences were not written in the negative review. In such cases, we labelled sentences (or phrases) to the right sentiment. Tables 1 and 2 give examples of assigning topics to raw reviews that consist of both positive and negative comments.

In this study, we randomly selected 191 out of 785 customer reviews. As a result of the topic labeling, we obtained 381 rows of customer reviews categorized by topics and sentiments (see Table 3).

4.2 Creation of Topic-Relevant Inference Rules in Hotel Reviews

This subsection focuses on annotating inference rules from multi-label text yielded from Step 1 (Definition 1). As aforementioned in Sect. 2, any inference rule is composed of two parts called the head and the body of the rule. Since all the reviews have been classified into eight topics and labelled by binary sentiments, we created topic-relevant inference rules by the following three steps.

Table 1. Examples of positive and negative reviews from a customer

Positive Review	Negative Review
The food was very tasty (I didn't have breakfast but ate dinner at the restaurant), staff very helpful and friendly, very nice bathroom, and the room was well equipped and also smelled very nice.	It was a bit loud outside but that was just normal for summertime I guess.

Table 2. Examples of assigning topics to the relevant text and sentiment

Assigned Topic	Topic-relevant Text	Sentiment
Food	The food was very tasty	Positive
Staff	staff very helpful and friendly,	Positive
Room	very nice bathroom, and the room was well equipped and also smelled very nice.	Positive
Room	It was a bit loud outside	Negative
Noise	(I didn't have breakfast but ate dinner at the restaurant), but that was just normal for summertime I guess.	Positive

1. Generated the head of each rule by the topic and the sentiment (cf. Subsection 3.3), resulting in 16 fixed literals, i.e., good_room, bad_room, good_staff, bad_staff, good_location, bad_location, good_food, bad_food, good_price, bad_price, good_facility, bad_facility, good_check-in, bad_check-in, good_ check-out, bad_check-out. This step was handled by an automatic program for the head's generation.
2. Generated the supporting literals in the body of each inference rule by translating from the topic-relevant text. The translation was handled by two human annotators with an agreed template. One annotator played the role of the leading annotator and the other annotator was the reviewer who ensured the consistency. Note that any inference rule has two primary parts: the *supporting literals* and the *contrary literals*. The used translation template for creating the supports was as follows:
   (a) Removed articles (viz. a, an, the),
   (b) Changed either a phrase or a sentence to the present simple tense,
   (c) Changed to lowercase letters,
   (d) Replace each space by an underscore ('_'),
   (e) Restricted to use the vocabularies appeared in the original text,
   (f) Restricted to maintain short phrases or sentences (i.e. without adjectives and adverbs) unless they were not understandable. For example, "Service was great" was changed to 'great_service' and "Very helpful host showed us how nearby bus takes to town-centre" was changed to 'helpful_host',

Table 3. Statistics of the ABA-HotelReviews Dataset for Task 1

La Veranda Hotel	Number of customer review
Original customer review	1500
Customer review that contains comments	785
Customer review that contains comments (randomly selected)	191
Topic-relevant Text	381 (Positive: 261, Negative: 120)

Table 4. Statistics of the ABA-HotelReviews Dataset for Task 2

Classified Topic	Number of customer review	Positive	Negative
Room	148	79	69
Staff	78	75	3
Location	72	53	19
Food	42	34	8
Price	12	4	8
Facility	20	5	15
Check-in	9	7	2
Check-out	0	0	0
Total	381	257	124

(g) Restricted to use noun phrases (if possible). For example, "The room was very clean" was changed to 'clean_room'.

3. Generated the contrary literals in the body of each inference rule by translating from each supporting literal and the topic-relevant text's sentiment (cf. Subsection 3.3). Indeed, there were two rules for creating contraries for positive and negative supports. For the positive support, we used the same word as presented in the supports but prefixed with 'no_evident_not_'. For the negative support, we prefixed it with 'have_evident_'. For example, if the text is "The restaurant area is nice and the staff very accommodating.", then the final generation contains four literals in the body of the inference rule as follows:
   (a) nice_restaurant_area (i.e. 'supporting literal')
   (b) accommodating_staff (i.e. 'supporting literal')
   (c) no_evident_not_nice_restaurant_area (i.e. 'contrary literal')
   (d) no_evident_not_accommodating_staff (i.e. 'contrary literal')

As a result, Table 4 summarizes the number of our annotated inference rules classified by all the eight topics (for both positive and negative sentiments). Note that in this study, our dataset contains zero row for the topic 'Check-out'. As always, all the annotations were agreed by three experts (two argumentation and one knowledge representation researchers) to ensure the appropriateness and applicability to use with argumentation reasoning engines in the future.

5 Experiments and Evaluations

Regarding our experiments for Tasks 1 and 2, we used the gpt-4o model from the OpenAI API[1] with temperature of 0 and top_p of 0.05. The API was configured with a low temperature and a low value for top_p to guarantee more predictable outputs (less creativity) and to control the vocabulary size in the generated outputs. Indeed, we considered only the top tokens to make the output as deterministic as possible. This configuration minimizes variability in the model's responses; thus it helps ensure the reproducibility.

In our experiments, each prompt was submitted five times. To ensure that the generated responses were not controversial, the results were selected based on the majority vote. If no majority responses appeared, we ran the prompt further (usually no more than nine times) until consistent responses could be generated. We designed prompts that could reflect and align with each task's objectives (see below).

Prompt 5.1

Please output the following [text] according to the **[Constraints]** in the **[Output Format]**.

[Constraints] The output should only be in the **[Output Format]**, and you must classify which part of the text corresponds to which Topic in the **[Topics]**.
Additionally, determine whether each classified element is Positive or Negative. If there is no corresponding element, put Null for both 'text' and 'label'. The most important constraint is not to include any extra characters such as newline characters, 'json', or backticks, or any other unnecessary text outside of the [output format].
If there are two or more elements of the same Topic, number each so that they do not conflict when converted to json format data. However, if they have the same NegPos label, keep them in one Text as much as possible.

[Topics] Room, Staff, Location, Food, Price, Facility, Check-in, Check-out

[Output Format]
{"Topics": {"Room":[{"text": "test","label": "Positive"}],
"Staff":[{"text": null,"label": null }],
"Location":[{"text": "test","label": "Positive"}],
"Food":[{"text": "test","label": "Negative"}],
"Price":[{"text": "test","label": "Positive"}],
"Facility":[{"text": "test","label": "Negative"}],
"Check-in":[{"text": "test","label": "Positive"}],
"Check-out":[{"text": null,"label": null}]}}

[Example]
"The room is big enough. But the room was a little bit dirty."
⇒ {"Topics":[{"Room1":[{"text": "The room is big enough.","label": "Positive"}],
"Room2":[{"text": "the room was a little bit dirty.","label": "Negative"}],
"Staff":[{"text": null,"label": null}],
"Location":[{"text": null,"label": null}],
"Food":[{"text": null,"label": null}],
"Price":[{"text": null,"label": null}],
"Facility":[{"text": null,"label": null}],
"Check-in":[{"text": null,"label": null}],
"Check-out":[{"text": null,"label": null}]}]}

[text]

[1] https://platform.openai.com.

5.1 Evaluation Metrics

Tasks 1 and 2 are kinds of information retrieval problems. Thus, it is inevitable to realize: **(1: micro-precision)** for each instance in the gold dataset, how precise can the GPT model respond to it?, and **(2: micro-recall)** for each instance in the gold dataset, how coverage can the GPT model deal with it? These two calculations are analogous to precision and recall, respectively; however, we attend to the scope of each individual review in the gold dataset instead. As these two measures give insights *locally*, it is natural to extend them for yielding *global* views. For this purpose, we also evaluate our implementations with two more aspects. **(3: average-precision)** on average, how precise can the model respond globally?, and **(4: average-recall)** on average, how coverage can the model respond globally?. These two values can be obtained by computing the average of all the micro-precisions and the average of all the micro-recalls, respectively.

5.2 Task 1: Creation of Multi-label Document Typesetting

Given 191 reviews, we executed 191 one-shot prompts independently to generate triples consisting of a topic, a topic-relevant text, and a sentiment corresponding to each input text. We designed one-shot prompt as illustrated in Prompt 4.2.1. As a result of its executions, we obtained 571 rows of topic-relevant text with 434 rows of positive, 135 rows of negative, and two rows of noises.

To evaluate the performance of our approach, we used the micro-precision and micro-recall as described earlier. Clearly, the total number of instances in the gold dataset and the generated responses are different. Therefore, these two metrics help measure "how precise and coverage can the model generate?" when compared with the gold dataset. Figures 3 and 4 present the total number of micro-precisions and micro-recalls. The responses achieved micro-precision of 1, 36.6%, micro-precision of 0.5, 12% and micro-precision of 0 is a little bit high at 16.8%. Furthermore, 63.4% achieved micro-recall of 1, while 16.8% achieved 0. For the global evaluation, Table 5 shows the averages of all the micro-precision and micro-recall's values for this task.

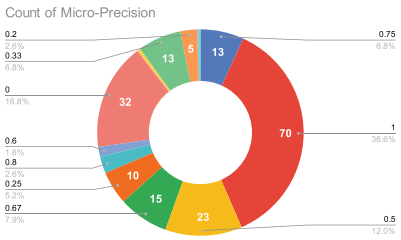

Fig. 3. Micro-precisions for Task 1

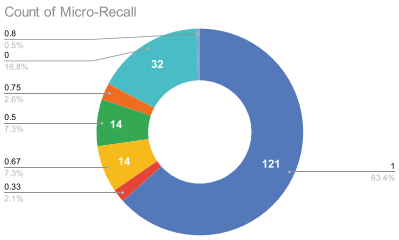

Fig. 4. Micro-recalls for Task 1

Prompt 5.2

Generate text in Assumption Based Argumentation (ABA) format from the **given text**. Use the following conditions carefully.

1. Claim is "good staff".
2. Supports are written in short words with no adjectives and no adverbs if it is not necessary for understanding.
3. For each support, use vocab from the original text. Do not provide synonyms. Check grammar. Do not provide further opinion.
4. Add a contrary for each support as a new support.
5. Each contrary must use the same word as presented in the support and each contrary starts with "no evident not".
6. Regarding the format of answer, provide a list of all supports and contraries. Do not separate supports and contraries into separated sections. Do not provide assumptions. Do not provide claims.

[**Example**] The staff were exceptional. So helpful and friendly. Went out of their way for us. ⇒ exceptional, helpful, friendly, went_out_of_their_way_for_us, no_evident_not_exceptional, no_evident_not_helpful, no_evident_not_friendly, no_evident_not_went_out_of_their_way_for_us

[**Text**] what will bring me back is the staff and the owners. Richard (the chef) is personally buying super fresh food for the restaurant. The owners have been super kind to me :)

5.3 Task 2: Creation of Topic-Relevant Inference Rules in Hotel Reviews

Task 2 is divided into two sub-tasks: (1) mining the rule's head and (2) mining the rule's body, in which the latter is also divided further: (3) mining the supporting literals and (4) mining the contrary literals.

For mining each rule's head, Subsect. 4.2 suggests an automatic procedure for the heads' generation. As this step did not involve the usage of any machine learning model, the evaluation metrics mentioned above were not used. Literally, given 381 labelled topics and their sentiments, our processing assigned the head directly based on the predefined 16 classes. For instance, the topic is 'room' and the sentiment is 'positive', then the head is 'good_room'. Otherwise, if the topic is 'room' and the sentiment is 'negative', then the head is 'bad_room'.

For mining each rule's body, given 381 topic-relevant texts, we executed 381 one-shot prompts (cf. Prompt 5.2) in accordance with the rules outlined in Subsect. 4.2. Note that some rules were not explicitly written in the prompt but were implicitly embedded in the one-shot's configuration.

To evaluate the performance of our approach, we compared all the 381 generated responses with the gold dataset and evaluate using the micro-precision and micro-recall. Figures 5 and 6 present these scores for all the individuals of each task, respectively. Indeed, 76.6% of the responses achieved micro-precision of 1, only 10.8% had micro-precision of 0.5. Furthermore, 88.2% achieved micro-recall of 1, while 3.7% had 0.67 micro-recall. Such high scores could be attributed by our relaxed threshold when comparing the responded text and the gold label in terms of structure, grammar, and semantic resemblance. Instead of requiring exact matches, we considered responses that conveyed

similar concepts as valid. Here are examples that were considered equivalently: helpful and helpful_host, slow_elevators and slow_elevator, room_wasn't_clean and not_clean_room, badly_soundproofed and bad_soundproof, and quality_television_image_poor and poor_quality_television. On the other hand, the following are examples that were considered inequivalently: fresh_handmade and fresh_breakfast, fresh_ handmade and handmade_breakfast, food and good_food, small_side_street and location_in_small_side_street, and took_time and took_time_to_explain. Table 5 shows the average of all the micro-precision and micro-recall's values.

In order to achieve better results, we plan to refine our prompts by incorporating more shots and use advanced prompt techniques in the future work.

Table 5. Average of the micro-precisions and micro-recalls

Task Number	Average of Micro-Precisions	Average of Micro-Recalls
Task 1	0.61	0.75
Task 2	0.88	0.94

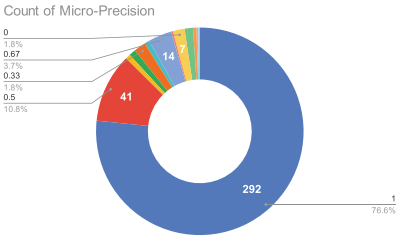

Fig. 5. Micro-precisions for Task 2

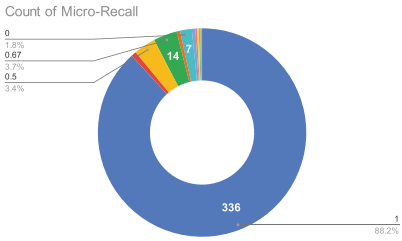

Fig. 6. Micro-recalls for Task 2

6 On Completion of ABA Mining: The Contrary Mining Task

As aforementioned in Sect. 2, this work adopts an extended ABA proposed by [14]. Thus, the notion of attack between arguments in ABA is defined in terms of the 'contraries' of assumptions, i.e., the set of assumptions is the set of assumptions from Task 2. The predicates beginning with 'no_evident_not' and 'have_evident_' are the mentioned assumptions.

Definition 3. *Let $\mathcal{L}$ be a language and $\mathcal{A}$ be a set of assumptions. We say that any* contrary mining *m is a function that maps from $\mathcal{A}$ into $2^{\mathcal{L}} \setminus \emptyset$.*

Like the above definitions, Definition 3 gives a general annotation framework for learning and evaluating the contrary mining task. For example, given that a

premise set $\mathcal{L}$ and an assumption set $\mathcal{A}$, an annotated corpus for this task can be in forms of a pair set $\{(x_i, y_i) \mid i \in \mathbb{N}\}$ where $x_i \in \mathcal{A}$ and $y_i \subseteq 2^{\mathcal{L}} \setminus \emptyset$.

A similar idea could be employed for the contrary mining task. For example, given the vocabulary obtained from Task 2, we could use multiple LLMs for annotating the contrary relations among all the propositions; these results will be evaluated by the human experts to justify the validity, resulting in another dataset. It remains however to show whether this methodology produces acceptable results in practice.

7 Related Work

As aforementioned, it was once stated by [32] that argument mining was coined out to connect the research results between natural language processing and knowledge representation, especially argumentation. However, most existing works focus on the philosophical understanding of argumentation from the viewpoint of machine learning. A precise and clear definition was given by [16], saying that argument mining is "the general task of analyzing discourse on the pragmatics level and applying a certain argumentation theory to model and automatically analyze the data at hand".

A recent survey from [30] further concluded that recent progress of NLP has influenced the argument mining research area. Indeed, automatic recognition and identification of arguments have been enabled in various domains and various machine learning models for the analysis and representation of argumentative structure are developed. So far, the most common argument components are recognized to include claims, premises, rebuttal, etc., following the well-known Toulmin's argumentation model [29,31]. According to Toulmin, arguments should be composed of facts that support a claim such that the claim is not made in an irresponsible manner. A claim is the conclusion of an argument. In order to establish such a claim, an individual needs 'data' to justify it. Similarly, in order to support the relation between the data to the claim, such individual needs at least a 'warrant'. There are further models of argumentation that motivate the research in argument mining. These include many forms of argumentation schemes. For instance, the argument forms that represent inferential structures of arguments introduced by [6], and the forms of inferencing patterns from premises to a conclusion in human's dialogue introduced by [25]. A variety of machine learning models are studied to deal with these argumentation structure, such as, Naive Bayes [22], RNNs [23], pre-trained language models [5], and other supervised learning techniques [13].

Various heterogeneous datasets have been proposed since the beginning of research in argument mining. As the field lacks of clear definitions on the notions of arguments due to the adoption of diverse argumentation models, the datasets thus far have been annotated from different definitions of argument components and of the relations holding between them. For example, in [27], a rhetorical-level analysis of scientific articles is introduced. Data annotation is based on the typical argumentation to be found in scientific articles. In [33], an argumentation

model and an annotation corpus for persuasive essays is studied following the work of [26], who define arguments' components for essays.

While there exist many progress on argument mining and philosophical models of argumentation, current results thus far are still not applicable to completely model structured argumentation frameworks in real world. A recent work by [24] investigated and showed that Toulmin's model of argumentation can be rewritten as a form of 2-tier argumentation system, where the first tier is Dung's abstract framework [11] and the second tier is any knowledge base which deduction is allowed. The main goal of [24] is to support the development of explainable AI systems in the real world by exploiting existing results in diverse areas of logic, e.g., formal argumentation, propositional logic, and the well-investigated proof procedures.

To the best of our knowledge, this is the first attempt that aims to fill a gap in connecting formal models of argumentation to natural language processing. We focus on the well-known ABA and introduce key machine learning tasks in order to transform raw data in the hotel reviews into an ABA framework.

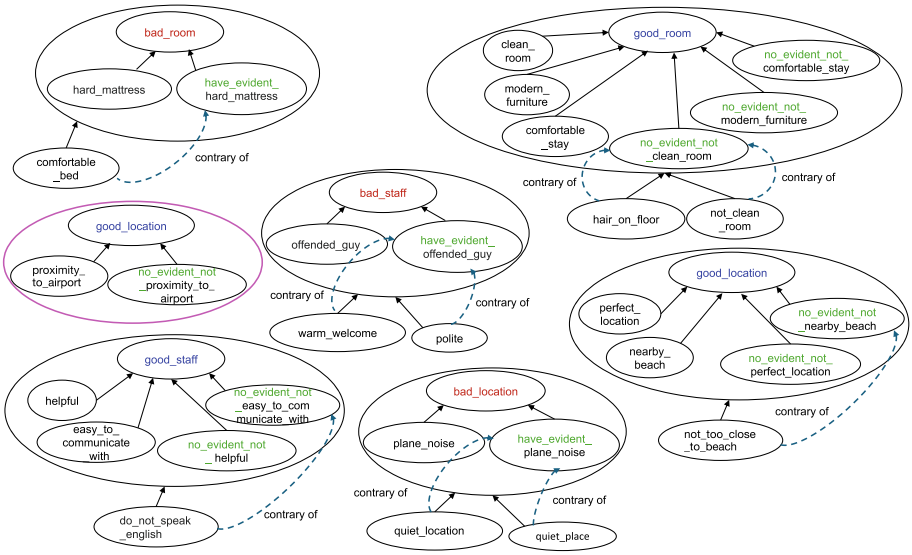

Fig. 7. A visualization of the extracted ABA knowledge base from the target dataset.

8 Conclusion

This paper introduces a novel research area called *assumption-based argumentation mining* (ABA mining) with our primary goal of connecting two mainstream areas in artificial intelligence: formal argumentation and natural language processing (NLP). To pursue this aim, we investigate hotel reviews in

Booking.com which contains arguments (positive and negative claims regarding customers' stays). Our analysis ensures that such data can be formalized in terms of ABA structure, allowing that the state-of-the-art ABA reasoners can be used to find further insights and gain understanding in the reviews. For example, by mining ABA structures from hotel reviews, one can realize which positive or negative claims are acceptable and explain why they are accepted or rejected.

Three novel tasks are formally defined and the dataset are annotated accordingly. We also experiment the applicability of our annotated corpus by leveraging the in-context learning capabilities of large language models in both Tasks 1 and 2. The findings show that in-context learning is promising. We aim to dive deeper into investigating further NLP techniques that could be useful for ABA mining.

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A Appendix A: Prompt Template

On the one hand, Prompt 4.2.1's template is static (i.e. an input to the prompt does not affect the template's content). On the other hand, the template of Prompt 5.2 could be dynamic in a sense that the content can be varied by the inputs. These inputs are parameters to the template, indicated by '{...}'.

Prompt A illustrates a parameterized content of the dynamic template of Prompt 5.2, where the parameters are 'good staff' and 'no evident not'. This template needs to be adjusted according to the 'rule's head'. Since the head is the 'claim' associated with each topic and sentiment, it is crucial to identify the specific claim used to mine its supports and contraries. Furthermore, the prefix used to the generated contraries also needs to be adjusted. This is to correct the meaning of generated contraries. For example,

- If the claim is 'good staff' (#1), the prefix will be 'no evident not' (#5).
- If the claim is 'bad staff' (#1), the prefix will be 'have evident' (#5).

B Appendix B: Additional Information

This section illustrates a graphical representation of our corpus for ABA mining. Figure 7 shows a visualization of the extracted ABA from the target dataset.

C Appendix C: Experiments and ABA-HotelReviews Datasets

Our experiments and ABA-HotelReviews datasets are accessible online at https://github.com/realearn-people/aba-mining-hotelreviews.

Prompt A: Dynamic Prompt Template
Generate text in Assumption Based Argumentation (ABA) format from the **given text**. Use the following conditions carefully.

1. Claim is {"good staff"}.
2. Supports are written in short words with no adjectives and no adverbs if it is not necessary for understanding.
3. For each support, use vocab from the original text. Do not provide synonyms. Check grammar. Do not provide further opinion.
4. Add a contrary for each support as a new support.
5. Each contrary must use the same word as presented in the support and each contrary starts with {"no evident not"}.
6. Regarding the format of answer, provide a list of all supports and contraries. Do not separate supports and contraries into separated sections. Do not provide assumptions. Do not provide claims.

[**Example**] The staff were exceptional. So helpful and friendly. Went out of their way for us. ⇒ exceptional, helpful, friendly, went_out_of_their_way_for_us, no_evident_not_exceptional, no_evident_not_helpful, no_evident_not_friendly, no_evident_not_went_out_of_their_way_for_us

[**Text**] what will bring me back is the staff and the owners. Richard (the chef) is personally buying super fresh food for the restaurant. The owners have been super kind to me :)

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